

Bitcoin Market Sentiment vs Trader Performance Analysis

Objective

The objective of this project is to analyze the relationship between Bitcoin market sentiment (Fear & Greed Index) and trader performance using Hyperliquid historical trading data.

Datasets Used

1. Fear & Greed Index Dataset

Columns:

- Date
- Classification (Fear, Greed, Extreme Fear, Extreme Greed, Neutral)

2. Historical Trader Dataset

Columns:

- Account
- Symbol
- Execution Price
- Side
- Size
- Closed PnL
- Leverage
- Timestamp

Methodology

1. Loaded both datasets using Pandas.
2. Converted date columns into datetime format.
3. Extracted trading dates from trader data.
4. Merged datasets using the date field.
5. Performed exploratory data analysis on:
 6. Closed PnL
 7. Trading Activity
 8. Market Sentiment
9. Generated visualizations and insights.

Key Findings

Profitability by Sentiment

The analysis shows that trader profitability varies across sentiment categories.

Average Closed PnL by Sentiment:

Sentiment	Average Closed PnL
Neutral	22.23
Extreme Greed	25.42
Fear	50.05
Greed	87.89

Observations

- The highest average trader profit was observed during Greed periods.
- Fear periods also produced positive profitability.
- Neutral sentiment showed the lowest profitability.
- Market sentiment appears to influence trader performance significantly.

Trading Behavior

- Traders tend to participate more actively during Greed phases.
- Positive market sentiment is associated with higher profitability.
- Sentiment indicators may provide useful signals for trading strategies.

Recommendations

1. Use market sentiment as an additional feature in trading decision-making.
2. Increase risk controls during periods of extreme sentiment.
3. Combine sentiment analysis with technical indicators for improved performance.
4. Monitor leverage usage during Greed periods to reduce excessive risk exposure.

Conclusion

This analysis demonstrates a meaningful relationship between Bitcoin market sentiment and trader performance. Traders generally achieved higher profitability during Greed periods compared to Neutral or Extreme Greed periods. Incorporating sentiment-based indicators into trading strategies may help improve decision-making and risk management.